

CE 405: Water and Wastewater Engineering

Course Code	CE 405
Course Name	Water and Wastewater Engineering
L-T-P-C	3-0-0-3
Prerequisites	Environmental Science (IC 230) or equivalent
Intended for	UG / M. Tech. /MS
Mutual Exclusion	None
Approval	52nd BoA (Nov 2023)

Course Contents

- **Planning and hydraulic design of water and sewerage system:** Population forecasting, equivalent sewage flow estimation, sewer pipelines and materials, sewer hydraulics, stormwater drainage, storm runoff estimation, sewer appurtenances, sewage pumping. (12 Lectures)
- **Water treatment processes:** Theory and application of water treatment process - aeration, coagulation, flocculation, sedimentation, filtration, and disinfection. (6 Lectures)
- **Wastewater treatment processes:** Unit operations and processes, selection of treatment processes, onsite sanitation, septic tank, grey water harvesting, primary treatment, screens, grit chamber, primary sedimentation tanks, construction, operation and maintenance aspects. (8 Lectures)
- **Secondary treatment processes:** Selection of treatment methods, hydraulic principles, activated sludge process and extended aeration systems, trickling filters and their design, sequencing batch reactor (SBR), membrane bioreactor (MBR), waste stabilization ponds, construction, operation and maintenance aspects. (10 Lectures)
- **Sewage Disposal and sludge management:** Standards for disposal methods, dilution, mass balance principle, self-purification of river, oxygen sag curve, deoxygenation and reaeration, dissolved oxygen modelling, reclamation and reuse of sewage, land disposal. agricultural uses of treated sewage, sludge characterization and sludge disposal methods. (6 Lectures)

Text Books

1. Mackenzie L. Davis, **Water and Wastewater Engineering: Design Principles and Practice**, McGraw-Hill Education, 2010.
2. Peavy, H.S., Rowe, D.R., Tchobanoglous, G., **Environmental Engineering**, McGraw-Hill Education, 2013

References

1. Metcalf, L., Eddy, H. P., & Tchobanoglous, G., **Wastewater engineering: treatment, disposal, and reuse** (Vol. 4), McGraw-Hill, 1979.
2. Karia. G. L., Christian, R. A., **Wastewater treatment: Concepts and design approach**, PHI Learning Pvt. Ltd., 2013.